



Melih Görgülü Data Scientist

📍 Berlin, Germany ✉ melihgorgulu98@gmail.com ☎ 0491636204605

🌐 [linkedin.com/in/melih-gorgulu](https://www.linkedin.com/in/melih-gorgulu) 🔗 melihgorgulu.github.io

Profile

I am a Data Scientist with strong experience in machine learning and predictive modeling. I deliver impactful projects by working in interdisciplinary teams, with a focus on both computer vision and LLM-based technologies. I'm skilled at solving complex challenges and translating data into actionable insights that drive decision-making. I combine strong academic credentials with practical, hands-on expertise, backed by a track record of impactful projects. More details are available on my personal website.

Professional Experience

Data Scientist, PAIR Finance

05/2025 – Present

Automated and optimized core business processes through production LLM systems and ML infrastructure improvements:

Berlin, Germany

- **Document Processing & Classification:** Built LLM-based classification pipelines that automated document routing and processing, enabling faster turnaround times and substantial cost savings.
- **System Optimization:** Redesigned a graph-based production pipeline's architecture, reducing computational GPU overhead by 30% and significantly improving system throughput.
- **ML Infrastructure & Deployment:** Managed the complete ML lifecycle—prompt engineering, model optimization, and production deployment of self-hosted LLMs.
- **GPU Resource Optimization:** Optimized computational resource allocation and model serving, balancing inference efficiency and system load for high-volume processing at scale.

Tech: SGLang, LangGraph, Docker, open-source LLMs, prompt engineering, model distillation, model fine-tuning.

Machine Learning Intern, AUDI

09/2024 – 03/2025

Contributed to generative AI projects, focusing on scalable deployment and model integration.

Ingolstadt, Germany

Completed Projects

REST API for Stable Diffusion Model: Designed and deployed a REST API using FastAPI for the Stable Diffusion deep learning model. Streamlined model deployment and scalability using AWS and Docker, with automation via GitHub Actions and automated tests.

LLM-Based License Compliance Checker: Integrated an LLM into Audi's AOSD process to detect and classify legally relevant modifications in FOSS license texts. Deployed a PostgreSQL Aurora DB for license text versioning and collaborated with the legal department using Agile/SCRUM methodologies.

AI Engineer Working Student, LINDE ENGINEERING

08/2022 – 10/2023

Enhanced AI-driven systems and RAG solutions, actively working with LLMs to improve functionality and user experience.

Munich, Germany

Completed Projects

Enhancement of RAG System with ChatGPT API: Improved RAG system functionality by integrating OpenAI's API and combining **ElasticSearch-based retrieval** with generative capabilities. Achieved a **12% improvement in F1 Score** by fine-tuning retrieval and generation workflows. Actively worked with **transformers**.

Locally Hosted AI Code Completion Extension for VSCode: Developed a local, offline code completion vscode extension using TypeScript, Hugging Face models and RESTful APIs to ensure data privacy.

Machine Learning Engineer - Full Time, ASELSAN
ASELSAN, Turkey's largest military defence company

12/2019 – 09/2021
Ankara, Turkey

- Accomplishments/Responsibilities**
- Enhanced the performance of a deep learning-driven detection system by achieving a **10% reduction in false alarm rate**.
 - Trained, analyzed, and optimized computer vision models and signal processing pipelines for detection systems.
 - Documented Software Requirements Specifications (**SRS**) and Software Test Descriptions (**STD**) to support project implementation and testing.

Education

Master's Degree, Technical University of Munich - Informatics
Graduated with merit. **Grade: 1.9/1.0**
Study Focus: **Machine Learning**
Thesis: Monocular Depth Prediction for Autonomous Driving, completed in Prof. Dr. Daniel Cremer's Lab

09/2021 – 10/2024
Munich, Germany

Bachelor's Degree, Gazi University - Computer Science
GPA: 3.76/4.0
Second-ranked degree in the Computer Science Department.

09/2016 – 05/2020
Ankara, Turkey

Skills

Programming Languages

- Python, C++

Frameworks/Technologies

- PyTorch, Numpy, Pandas
- Sglang, LangGraph, VLLM
- FastAPI, Pytest
- AWS, CI/CD Tools, Docker

Technical Knowledge

- Production ML Systems
- Computer Vision, LLM's, Transformers
- Multi View Geometry
- Software Engineering, Design Patterns

Databases

- PostgreSQL, MongoDB

Publications

- DynaMoN**
- Our paper **has been accepted** to the **2025 IEEE Robotics and Automation Letters (RA-L)**
 - Paper Link:** <https://ieeexplore.ieee.org/document/10777295>
 - Github Page:** <https://hannahhaensen.github.io/DynaMoN/>

Languages

Turkish Native	English Fluent	German B1
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